Predicting U.S. Flight Departure Delays

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Introduction

There are hundreds of thousands of flights every day on average and flight delays affect connections, staffing, and travel plans. In this study our group has decided to look at the relationship between certain input variables like weather and origin and how they can affect the output variable which in this case is flight delay/cancellation. Our main question is how well information available one hour before departure can identify domestic U.S. flights likely to leave at least 15 minutes late. We also want to learn which flight, schedule, and weather conditions are associated with delays. These results could help travellers and airport staff plan for disruption.

For this our group has found a dataset that combines domestic flights from across the United States of America from 2023 to 2025 and compares their input variables to whether or not the flight was delayed or cancelled. Furthermore, if the flight was solely delayed and not cancelled how long was that delay. This dataset only looks at domestic U.S. flights but it does not cover all carriers. Prediction and explanation serve different purposes here. We will describe associations among the variables and compare forecasting approaches, but the data cannot establish that any one condition causes a delay.

Data

Our github.com/JohnPierman/US-Flight-Delay-Study documents the prepared dataset and its reproducible code. The locally generated dataset combines 2023–2025 Bureau of Transportation Statistics (BTS) reporting-carrier domestic flights with airport details and archived Iowa Environmental Mesonet weather. The source contains 20,928,579 scheduled-flight records. After excluding cancellations and records without a clock-consistent departure-delay label, **20,420,136** flights remain for the main analysis.

The main outcome is whether a flight departed at least 15 minutes late. We will also examine delay in minutes to understand its distribution. Available predictors include airline, airports, scheduled departure time, route distance, weather observed before the one-hour cutoff, and forecasts issued by that cutoff. Actual flight outcomes will not be used to predict the same flight.

Exploratory delay summary	
Mean	12.8 min
Median	−2.0 min
90th percentile	44.0 min
Departed early (< 0 min)	58.0%
At least 15 min late	20.9%

Table 1: Eligible domestic flights. Source: project repository.

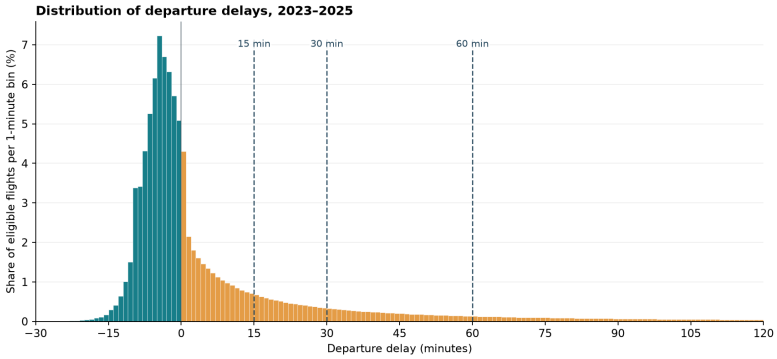


Figure 1: Each bar spans one minute. Dashed lines mark 15-, 30-, and 60-minute delays. The plot omits 0.005% below −30 min and 2.99% at 120 min or more.

About one in five eligible flights left at least 15 minutes late. The long right tail makes a normal curve a poor description of delay times. Weather coverage is uneven: 73.3% of flights have a matched origin observation, and 37.2% have a bounded destination forecast. We will account for missing values and check whether exclusions change the mix of airports, airlines, or seasons. The data omit some carriers and all international flights. Final schedules and weather timestamps also leave some uncertainty about what was available at the prediction cutoff.

Exploratory patterns

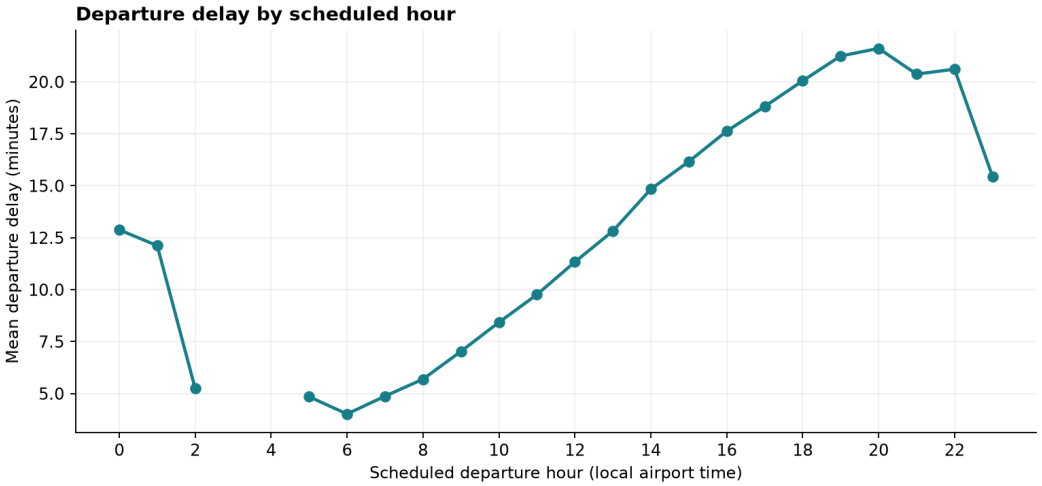


Figure 2: Mean delay rises through much of the day. Hours with fewer than 1,000 flights are omitted.

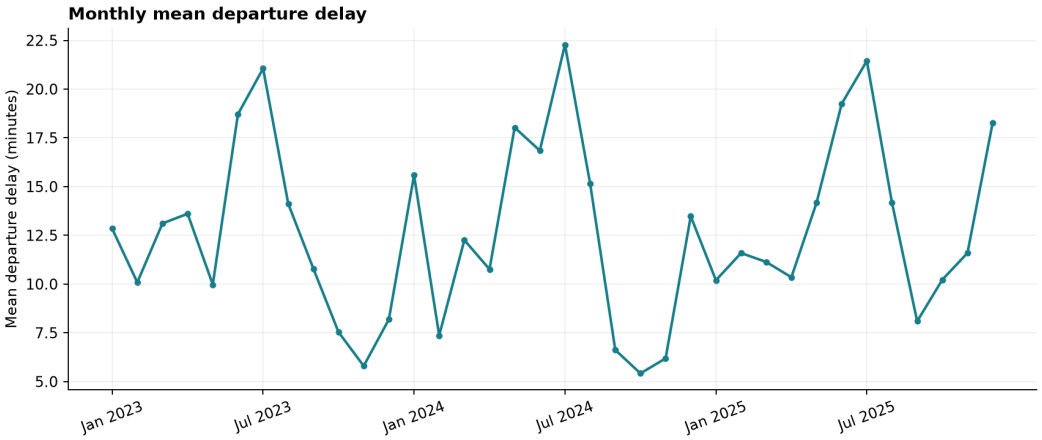


Figure 3: Monthly mean delay varies over the three years, with several summer peaks.

These plots suggest that scheduled hour and season deserve close attention. They show unadjusted averages, so the patterns could also reflect changes in routes, airports, carriers, or weather. The final analysis will examine those covariates together rather than attribute the pattern to a single factor.

Proposed analysis

Relationships among variables. We will compare delay rates across airports, airlines, routes, departure hours, and seasons. We will also check how weather measures relate to one another and whether their links with delay change by season or airport. Comparisons within the same season will help show where an apparent relationship reflects the mix of flights in each group.

Prediction. We will compare a linear model with a nonlinear model drawn from course methods. Validation within 2023–2024 will guide model complexity and the treatment of missing values. The held-out 2025 flights will provide a final check of accuracy and of performance across airports and seasons. Keeping season in the analysis will help us separate recurring calendar patterns from longer-term changes.

Additional predictors. Holiday indicators and scheduled airport traffic may add useful context if they can be defined using information available by the cutoff.

Cancellation comparison. We may obtain a comparable international-flight sample to describe cancellation rates alongside the domestic master records. Cancellations are excluded from the main delay outcome, and international flights would remain a separate descriptive comparison. We will check that coverage and definitions are comparable before drawing conclusions.